

◆ 1. Variables

- **Definition:** A variable stores data that can change.
- **Examples from your code:**

```
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secret_number = random.randint(1, 10)
guess = None
```

- `secret_number` holds the number the user must guess.
 - `guess` stores the user's input.
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◆ 2. Data Types

- **int (Integer):** Whole numbers (e.g., `1`, `10`).
- **str (String):** Text input from user (e.g., `"Enter your guess"`).
- **bool (Boolean):** Used implicitly for conditions (`True` or `False`).

Example in your code:

```
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guess = int(guess) # converting string input to integer
```

◆ 3. Decision Making (If-Else Statements)

- Used to control the flow of the program based on conditions.

Example in your code:

```
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if not guess.isdigit():
    print("Please enter a valid number.")
elif guess < 1 or guess > 10:
```

```
    print("Number must be between 1 and 10.")
elif guess < secret_number:
    print("Too low!")
elif guess > secret_number:
    print("Too high!")
else:
    print("Correct! You win!")
```

◆ 4. Loops (While Loop)

- Repeats a block of code until a condition is met.

Example in your code:

```
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while True:
    guess = input("Enter your guess: ")
    ...
    if guess == secret_number:
        break
```

- The loop runs until the correct number is guessed (**break** stops it).
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◆ 5. Functions

- **Built-in functions used in your code:**
 - `print()` → Displays messages.
 - `input()` → Gets user input.
 - `int()` → Converts string to integer.
 - `random.randint()` → Generates a random number.

Example:

```
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```

```
secret_number = random.randint(1, 10)
```

◆ 6. Lists (Optional Enhancement)

- **Definition:** A collection of items.
- While not in your current code, you could store guesses like this:

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```
guesses = []          # Create an empty list
guesses.append(guess) # Add each guess to the list
```